🚀 RevOps Pipeline Health & Sales Performance Dashboard

A hands-on GTM analytics project built in Databricks – By Flora LEE

Objective
Transform multi-table CRM data into actionable pipeline and productivity insights for Sales leadership.

CRM data modelling

SQL & transformation

Pipeline analysis

Sales Performance

RevOps reporting

Dashboard development

Tools: Databricks SQL • Unity Catalog • Delta tables • Python/PySpark • Dashboarding

1. Project Overview

Project scope

Build an end-to-end **RevOps analytics workflow** using a sample CRM dataset to turn opportunity-level data into actionable insights on pipeline health, sales performance and opportunity concentration.

Dataset

- Source: **Kaggle — CRM Sales Predictive Analytics**
- Sample B2B **computer hardware sales** CRM dataset
- Includes opportunity, account, product and sales-team information
- Analysed using **Databricks, SQL and PySpark**

Project covers

1. CRM data ingestion and data modelling
2. Data quality and data preparation
3. Pipeline analysis by stage, rep, product and account
4. Open opportunity concentration
5. Win rate and sales cycle analysis
6. Pipeline aging and sales productivity
7. Leadership dashboarding

Dataset limitations

Dataset Limitations

- This is a **sample CRM dataset**, so the analysis should be interpreted as a portfolio demonstration rather than a real operating forecast.
- ***close_value*** is primarily available for **closed opportunities**, limiting dollar-value analysis of open pipeline.
- No **quota / target** field → rep attainment and pipeline coverage cannot be calculated.
- No historical **stage-change timestamps** → true stage-to-stage conversion cannot be measured.
- No probability field → weighted forecasting would require external assumptions.
- No activity data such as calls, emails or meetings → activity-based productivity cannot be assessed.
- Historical/sample data → insights demonstrate the analytical approach rather than current commercial performance.

Source:
https://www.kaggle.com/datasets/agungpambudi/crm-sales-predictive-analytics?utm_source=chatgpt.com

2. Data Set & Architecture

CRM-style data model used for the project

sales_pipeline.csv

Core opportunity-level fact table
opportunity, rep, product, account, stage, dates, value

accounts.csv

Account master data
Used for account-level segmentation and enrichment

products.csv

Product master data
Used for product mix and pipeline analysis

sales_teams.csv

Sales hierarchy / rep context
Used for productivity and attainment views

Core sales_pipeline fields identified

opportunity_id	sales_agent	product	account
deal_stage	engage_date	close_date	close_value

3. Project Flow

From raw CRM files to leadership-ready RevOps insight

01 **Define variables**
Catalog, schema, volume

02 **Create volume**
Unity Catalog storage

03 **Upload CRM data**
CSV source files

04 **Load pipeline**
Create DataFrame

05 **Create SQL view**
Python → SQL bridge

06 **Clean data**
Dates, numbers, quality

07 **Stage analysis**
Pipeline by stage

08 **Pipeline metrics**
Health & productivity

09 **Dashboard**
Leadership reporting

10 **Recommendations**
Actions & next steps

4. Business Metrics — Opportunity & Pipeline Analytics

Opportunity by Stage

SQL: COUNT opportunities + GROUP BY deal stage.
Observation: identify where opportunity is concentrated and whether late-stage pipeline is sufficient.

SQL

```
SELECT deal_stage,
       COUNT(*) AS
opportunity_count
FROM
flora_workspace.default.sales
_pipeline
GROUP BY deal_stage
ORDER BY opportunity_count
DESC;
```

	deal_stage	opportunity_count
1	Won	4238
2	Lost	2473
3	Engaging	1589
4	Prospecting	500

Closed Won by Rep

SQL: GROUP BY sales_agent.
Observation: compare pipeline ownership, concentration and whether some reps have materially lower coverage.

SQL

```
SELECT sales_agent,
       COUNT(*) AS won_opportunities,
       CONCAT('$',
FORMAT_NUMBER(SUM(close_value), 0)) AS
amount
FROM flora_workspace.default.sales_pipeline
WHERE deal_stage = 'Won'
GROUP BY sales_agent
ORDER BY SUM(close_value) DESC
```

	sales_agent	won_opportunities	amount
1	Darcel Schlecht	349	$1,153,214
2	Vicki Laflamme	221	$478,396
3	Kary Hendrixson	209	$454,298
4	Cassey Cress	163	$450,489
5	Donn Cantrell	158	$445,860
6	Reed Clapper	155	$438,336
7	Zane Levy	161	$430,068
8	Corliss Cosme	150	$421,036
9	James Ascencio	135	$413,533
10	Daniell Hammack	114	$364,229
11	Maureen Marcano	149	$350,395
12	Gladys Colclough	135	$345,674
13	Markita Hansen	130	$328,792

Closed Won by product

SQL: GROUP BY product.
Observation: identify product mix, concentration and whether higher-value products dominate the forecast.

SQL

```
SELECT product,
       COUNT(*) AS opps,
 CONCAT('$',
FORMAT_NUMBER(SUM(close_value), 0)) AS
amount
FROM
flora_workspace.default.sales_pipeline
GROUP BY product
ORDER BY amount DESC;
```

	product	opportunity_count	amount
1	GTX Plus Basic	1383	$705,275
2	GTX Basic	1866	$499,263
3	MG Special	1651	$43,768
4	GTK 500	40	$400,612
5	GTXPro	1480	$3,510,578
6	GTX Plus Pro	968	$2,629,651
7	MG Advanced	1412	$2,216,387

Average deal size

SUM(close_value) / COUNT(opportunity_id).
Observation: compare average deal size by understand mix and productivity.

SQL

```
%sql

SELECT
    CONCAT(
        '$',
        FORMAT_NUMBER(
            AVG(close_value),
            0
        )
    ) AS average_won_deal_size

FROM
flora_workspace.default.sales_pipeline

WHERE deal_stage = 'Won'
  AND close_value IS NOT NULL;
```

	average_won_deal_size
1	$2,361

4. Business Metrics — Opportunity & Pipeline Analytics

Open Opportunity Concentration by Account

Rank accounts by # of open pipeline and calculate share of total.

SQL

```
WITH account_opportunities AS (
    SELECT
        account,
        COUNT(*) AS open_opportunity_count
    FROM flora_workspace.default.sales_pipeline
    WHERE deal_stage NOT IN ('Won', 'Lost')
      AND account IS NOT NULL
    GROUP BY account
),

total_open AS (
    SELECT
        SUM(open_opportunity_count) AS total_open_opportunities
    FROM account_opportunities
)

SELECT
    account,
    open_opportunity_count,

    CONCAT(
        ROUND(
            100.0 * open_opportunity_count / total_open_opportunities,
            2
        ),
        '%'
    ) AS opportunity_share_pct,

    RANK() OVER (
        ORDER BY open_opportunity_count DESC
    ) AS opportunity_rank

FROM account_opportunities
CROSS JOIN total_open

ORDER BY opportunity_rank;
```

	account	open_opportunity_count	opportunity_share...
1	Donquadtech	14	2.11%
2	Betasoloin	14	2.11%
3	Inity	13	1.96%
4	Iselectrics	13	1.96%
5	Blackzim	12	1.81%
6	Faxquote	12	1.81%
7	Plussunin	12	1.81%
8	Zoomit	12	1.81%
9	Dontechi	12	1.81%
10	Condax	11	1.66%
11	Hatfan	11	1.66%
12	Xx-zobam	11	1.66%
13	Bluth Company	11	1.66%
14	Dalttechnology	11	1.66%
15	Sumace	11	1.66%

4. Business Metrics — Opportunity & Pipeline Analytics

Open Opportunity Concentration by Rep

Rank Rep by # of open pipeline and calculate share of total.

SQL

```sql
WITH rep_opportunities AS (
    SELECT
        sales_agent,
        COUNT(*) AS open_opportunity_count
    FROM flora_workspace.default.sales_pipeline
    WHERE deal_stage NOT IN ('Won', 'Lost')
      AND sales_agent IS NOT NULL
    GROUP BY sales_agent
),

total_open AS (
    SELECT
        SUM(open_opportunity_count) AS total_open_opportunities
    FROM rep_opportunities
)

SELECT
    sales_agent,
    open_opportunity_count,

    CONCAT(
        ROUND(
            100.0 * open_opportunity_count / total_open_opportunities,
            2
        ),
        '%'
    ) AS opportunity_share_pct,

    RANK() OVER (
        ORDER BY open_opportunity_count DESC
    ) AS opportunity_rank

FROM rep_opportunities
CROSS JOIN total_open

ORDER BY opportunity_rank;
```

	sales_agent	open_opportunity_count	opportunity_share_pct
1	Darcel Schlecht	194	9.29%
2	Anna Snelling	112	5.36%
3	Vicki Laflamme	104	4.98%
4	Kary Hendrixson	103	4.93%
5	Versie Hillebrand	97	4.64%
6	Kami Bicknell	90	4.31%
7	Zane Levy	88	4.21%
8	Marty Freudenburg	87	4.16%
9	Cassey Cress	85	4.07%
10	Gladys Colclough	85	4.07%
11	Corliss Cosme	81	3.88%
12	Jonathan Berthelot	81	3.88%
13	Lajuana Vencill	80	3.83%
14	Markita Hansen	79	3.78%
15	Maureen Marcano	72	3.45%

4. Business Metrics — Opportunity & Pipeline Analytics

Average Sales Cycle by Rep

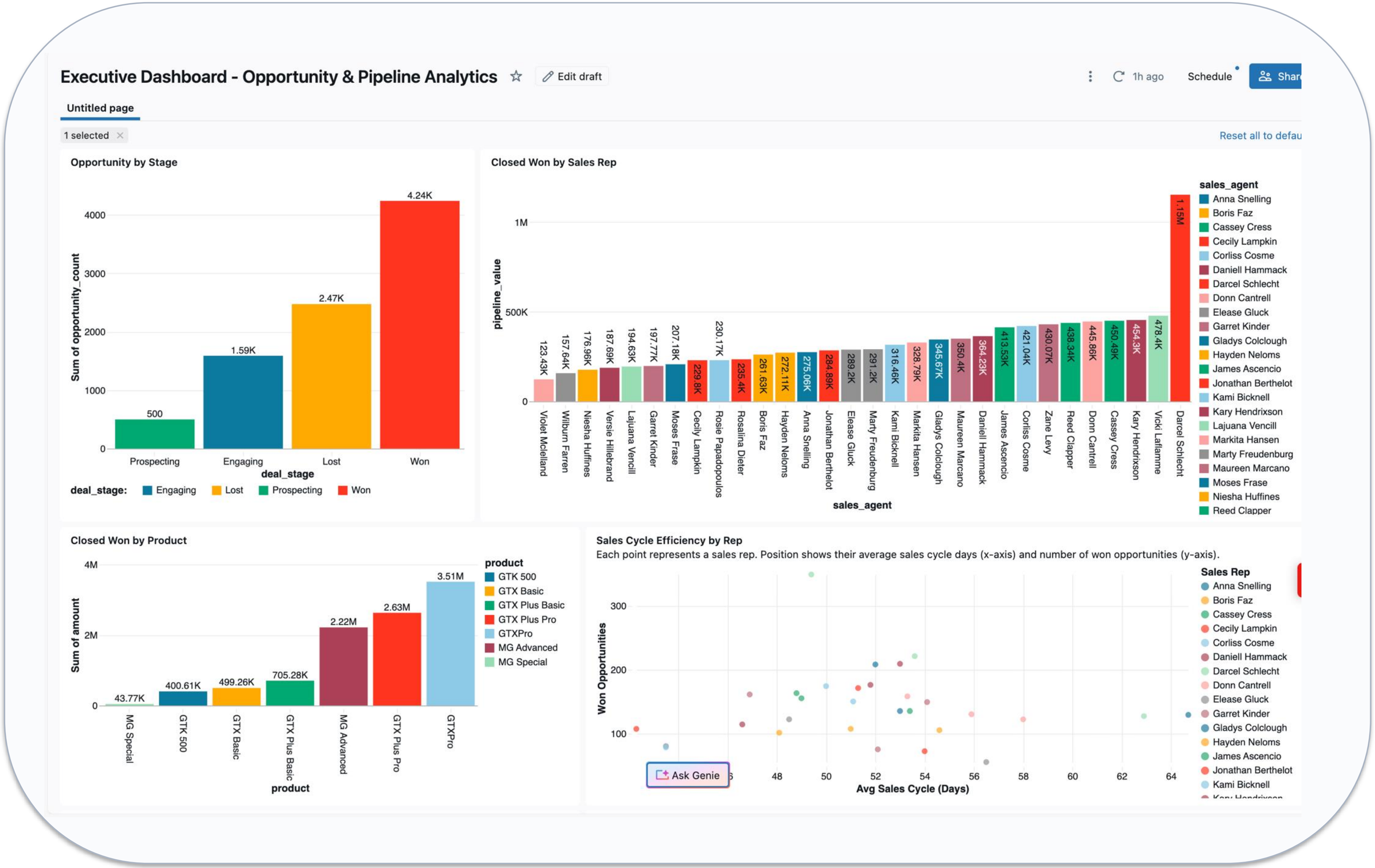
Identify reps or segments where opportunities are taking longer to close

SQL

```
SELECT
    sales_agent,

    COUNT(*) AS closed_opportunities,

    ROUND(
        AVG(
            DATEDIFF(close_date, engage_date)
        ),
        1
    ) AS avg_sales_cycle_days

FROM flora_workspace.default.sales_pipeline

WHERE deal_stage IN ('Won', 'Lost')
  AND engage_date IS NOT NULL
  AND close_date IS NOT NULL

GROUP BY sales_agent

ORDER BY avg_sales_cycle_days DESC;
```

	sales_agent	won_opportunities	avg_sales_cycle_days
1	Moses Frase	129	64.7
2	Lajuana Vencill	127	62.9
3	Violet Mclelland	122	58
4	Wilburn Farren	55	56.5
5	Markita Hansen	130	55.9
6	Niesha Huffines	105	54.6
7	Maureen Marcano	149	54.1
8	Rosalina Dieter	72	54
9	Vicki Laflamme	221	53.6
10	James Ascencio	135	53.4
11	Donn Cantrell	158	53.3
12	Gladys Colclough	135	53
13	Kary Hendrixson	209	53
14	Garret Kinder	75	52.1
15	Anna Snelling	208	52

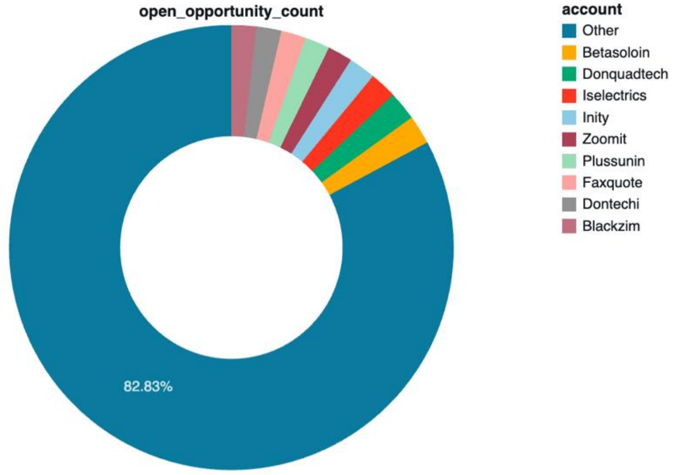
5. Dashboard — Opportunity & Pipeline Analytics



5. Dashboard — Opportunity & Pipeline Analytics

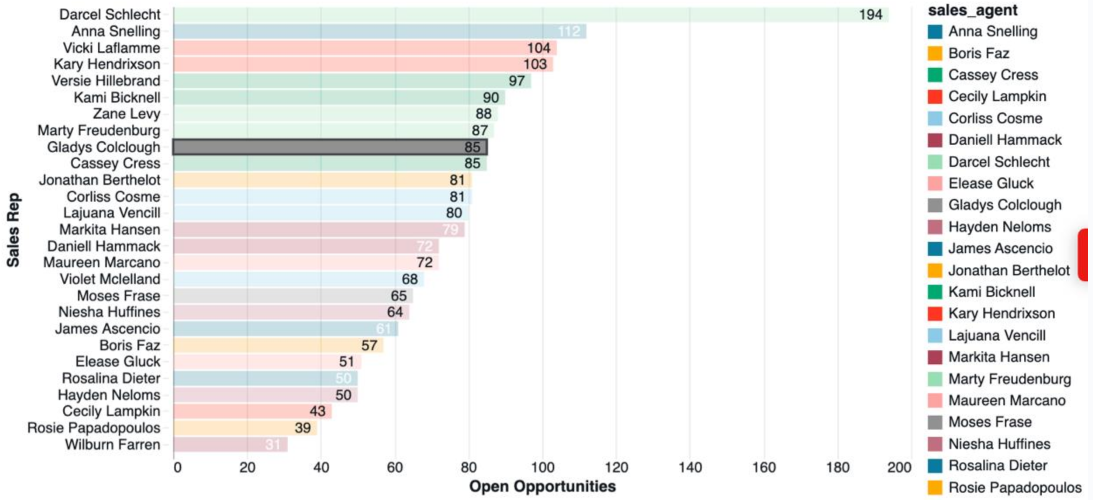
Open Opportunity Concentration by Account

The top 9 accounts represent 17.17% of all open opportunities, indicating that active opportunity volume is not concentrated among a relatively small group of accounts.



Open Opportunity Concentration by Sales Rep

Distribution of active opportunities across sales representatives, ordered by opportunity count.



5. Dashboard – Target Audience

C-level

How healthy and concentrated is the sales opportunity base?

Sales Leadership

Which reps and accounts have the most active opportunities?

Sales Managers

Where are opportunities getting stuck and which reps may need attention?

Marketing Leadership

Which products/accounts represent the strongest opportunity volume?

RevOps / Sales Ops

Where are data-quality, pipeline and process issues?

Account Executives

Which opportunities/accounts require prioritisation?

6. Other SaaS Business Metrics — Funnel, Aging & Health

Stage conversion

Identify funnel leakage and stages with the biggest drop-off.

Win rate

Assess conversion and compare by rep, product or period.

Sales cycle

Spot long-cycle segments and forecast slippage.

Pipeline aging

Flag stale opportunities requiring rep action or cleanup.

Pipeline coverage

Assess whether pipeline is large enough to support the target.

Forecast variance

Test whether the forecast model is optimistic or pessimistic.

7. Business Insights

1. Overall

Opportunity volume is relatively well distributed across accounts, while closed-won revenue is more concentrated across a small number of products. Sales-cycle differences across reps also highlight areas where Sales leadership can investigate potential process or deal-complexity differences.

2. Opportunity Health

48.2% of opportunities are Won, while **23.7% remain open**. Engaging represents the majority of active opportunity volume, suggesting most open opportunities have progressed beyond initial prospecting.

3. Concentration Risk

The **top 9 accounts represent only 17.17%** of open opportunities, indicating low account concentration. Rep concentration is higher: the top 5 reps own approximately **29.2%** of open opportunity volume.

4. Product Revenue Mix

Closed-won revenue is highly concentrated: **GTXPro, GTX Plus Pro and MG Advanced contribute ~83.5%** of closed-won revenue, creating significant dependency on these product lines.

5. Sales Efficiency

Average sales cycle varies across reps, ranging from approximately **52 to 64.7 days** among the reps shown. Longer cycles should be investigated alongside deal complexity and customer mix rather than treated as a standalone performance issue

6. RevOps Data Quality

Missing CRM fields — particularly account and date/value fields — highlight the importance of **data-quality controls** for reliable pipeline and sales reporting.